

Chapter 1

Table 1-3 summarizes the key points about how adjacent layers work together on a single computer and how one layer on one computer works with the same networking layer on another computer.

Table 1-3 Summary: Same-Layer and Adjacent-Layer Interactions

Concept	Description
Same-layer interaction on different computers	The two computers use a protocol to communicate with the same layer on another computer. The protocol defines a header that communicates what each computer wants to do.
Adjacent-layer interaction on the same computer	On a single computer, one lower layer provides a service to the layer just above. The software or hardware that implements the higher layer requests that the next lower layer perform the needed function.

Figure 1-10 repeats the familiar case in which web server Larry wants to send part of a web page to Bob, but now with details related to IP. On the lower left, note that server Larry has the familiar application data, HTTP header, and TCP header ready to send. In addition, the message now contains an IP header. The IP header includes a source IP address of Larry's IP address (1.1.1.1) and a destination IP address of Bob's IP address (2.2.2.2).

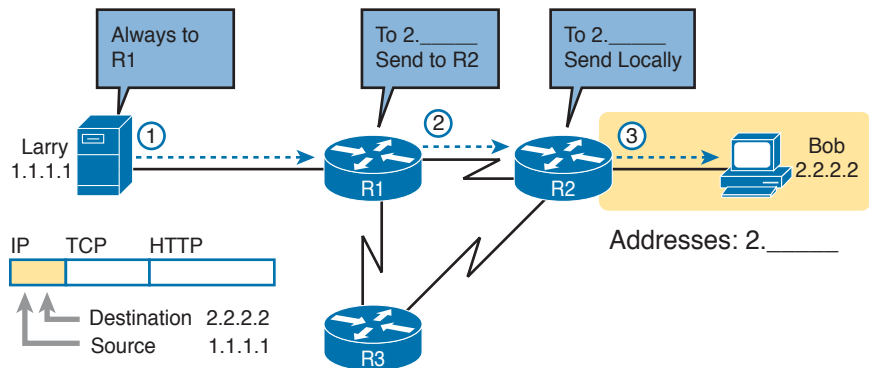


Figure 1-10 *Basic Routing Example*

Because each layer provides a service to the layer above it, take a moment to think about the IP logic related to Figure 1-10. In that example, host Larry's IP logic chooses to send the IP packet to a nearby router (R1). However, while Figure 1-10 shows a simple line between Larry and router R1, that drawing means that some Ethernet LAN sits between the two. Figure 1-11 shows four steps of what occurs at the link layer to allow Larry to send the IP packet to R1.

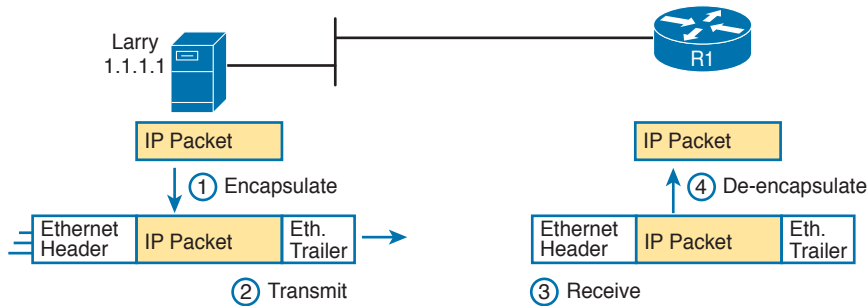


Figure 1-11 *Larry Using Ethernet to Forward an IP Packet to Router R1*

The numbers in Figure 1-12 correspond to the five steps in this list, graphically showing the same concepts. Note that because the application layer often does not need to add a header, the figure does not show a specific application layer header, but the application layer will also at times add a header as well.

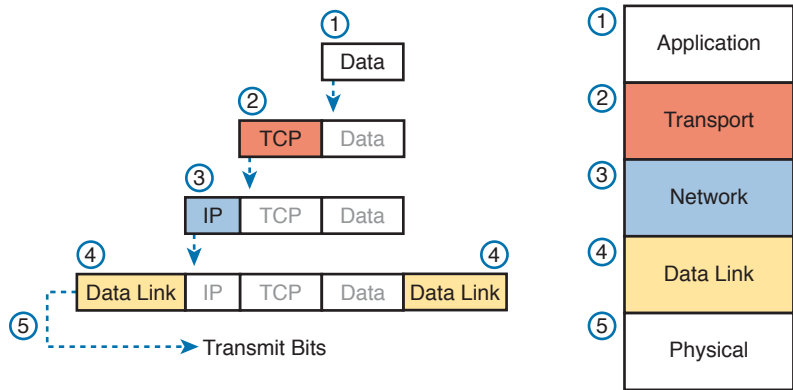


Figure 1-12 *Five Steps of Data Encapsulation: TCP/IP*

One reason this chapter takes the time to show the encapsulation steps in detail has to do with terminology. When talking and writing about networking, people use *segment*, *packet*, and *frame* to refer to the messages shown in Figure 1-13 and the related list. Each term has a specific meaning, referring to the headers (and possibly trailers) defined by a particular layer and the data encapsulated following that header. Each term, however, refers to a different layer: segment for the transport layer, packet for the network layer, and frame for the link layer. Figure 1-13 shows each layer along with the associated term.



Figure 1-13 *Perspectives on Encapsulation and “Data”**

* The letters LH and LT stand for link header and link trailer, respectively, and refer to the data-link layer header and trailer.

Today, the OSI model can be used as a standard of comparison to other networking models. Figure 1-14 compares the seven-layer OSI model with both the four-layer and five-layer TCP/IP models.

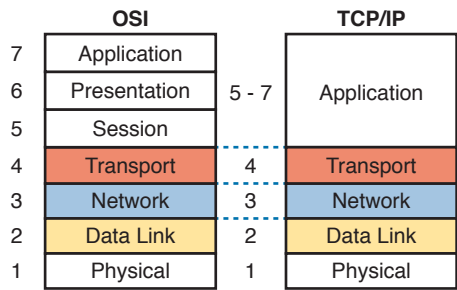


Figure 1-14 *OSI Model Compared to the Two TCP/IP Models*

OSI uses a more generic term to refer to messages, rather than frame, packet, and segment. OSI uses the term *protocol data unit* (PDU). A PDU represents the bits that include the headers and trailers for that layer, as well as the encapsulated data. For example, an IP packet, as shown in Figure 1-13, using OSI terminology, is a PDU, more specifically a *Layer 3 PDU* (abbreviated L3PDU) because IP is a Layer 3 protocol. OSI simply refers to the Layer x PDU (L x PDU), with x referring to the number of the layer being discussed, as shown in Figure 1-15.

L#H - Layer # Header
L#T - Layer # Trailer

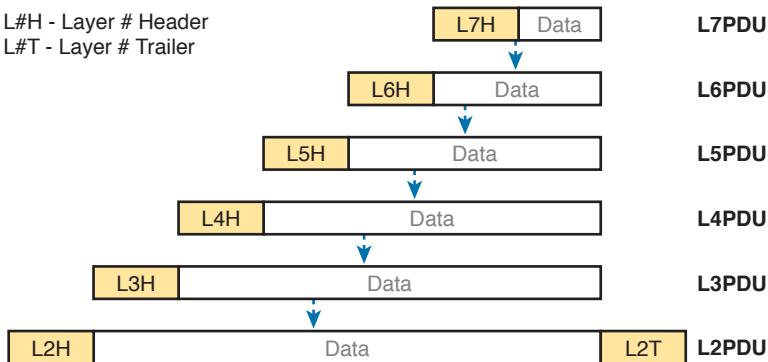


Figure 1-15 OSI Encapsulation and Protocol Data Units